Approximate Functions

VLC = Vision Learning Control

Going Deep

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Overview

- No free lunch and universal approximation
- Why go deep?
- Problems of going deep
- Some fixes:
  - Improving gradient flow with skip connections
  - Regularising with Dropout

No Free Lunch

- Statistical learning theory claims that a machine can generalise well from a finite training set.
- This contradicts basic inductive reasoning which says to derive a rule describing every member of a set one must have information about every member.
- Machine learning avoids this problem by learning probabilistic[1] rules which are *probably* correct about *most* members of the set they concern.
- But, **no free lunch theorem** states that every possible classification machine has the *same error* when averaged over *all possible* data-generating distributions.
  - **No machine learning algorithm is universally better than any other!**
  - Fortunately, in the real world, data is generated by a small subset of generating distributions...

[1] or perhaps more generally rules which are not certain

The Universal Approximation Theorem

Let $\psi : \mathbb{R} \to \mathbb{R}$ be a nonconstant, bounded, and continuous function. Let I_m denote the m-dimensional unit hypercube $[0, 1]^m$. The space of real-valued continuous functions on I_m is denoted by $C(I_m)$. Then, given any $\varepsilon > 0$ and any function $f \in C(I_m)$, there exists an integer N, real constants $v_i, b_i \in \mathbb{R}$ and real vectors $w_i \in \mathbb{R}^m$ for $i = 1, \ldots, N$, such that we may define:

$F(x) = \sum_{i=1}^{N} v_i \psi(w_i^T x + b_i)$ as an approximate realization of the function f ; that is,

$|F(x) - f(x)| < \varepsilon \; \forall \; x \in I_m.$

$\implies$ simple neural networks can represent a wide variety of interesting functions when given appropriate parameters.

So a single hidden layer can approximate any continuous function?

- Yes!
- But, ...
  - to get the precision you require (small ε), you might need a really large number of hidden units (very large N).
  - worse-case analysis shows it might be exponential (possibly one hidden unit for *every* input configuration)
  - We've not said anything about learnability...
    - The optimiser might not find a good solution[2].
    - The training algorithm might just choose the wrong solution as a result of overfitting.
    - *There is no known universal procedure for examining a set of examples and choosing a function that will generalise to points out of the training set.*

[2] note that it has been shown that the gradients of the function are approximated by the network to an arbitrary precision

Then Why Go Deep?

- There are functions you can compute with a deep neural network that shallow networks require exponentially more hidden units to compute.
  - The following function is more efficient to implement using a deep neural network: $y = x_1 \oplus x_2 \oplus x_3 \oplus \cdots \oplus x_n$
- We should care about the data generating distribution (c.f. NFL).
  - Real-world data has significant structure; often believed to be generated by (relatively) simple low-dimensional latent processes.
  - Implies a prior belief that the underlying factors of variation in data can be explained by a hierarchical composition of increasingly simple latent factors
- Alternatively, one could just consider that a deep architecture just expresses that the function we wish to learn is a program made of multiple steps where each step makes use of the previous steps outputs.
- **Empirically, deeper networks just seem to generalise better!**

What are the problems?

- Learnability is still hard
  - Problems of gradient flow
  - Horrible symmetries in the loss landscape
  - Overfitting

Vanishing and Exploding Gradients

- The vanishing and exploding gradient problem is a difficulty found in training NN with gradient-based learning methods and backpropagation.
- In training, the gradient may become vanishingly small (or large), effectively preventing the weight from changing its value (or exploding in value).
- This leads to the neural network not being able to train.
- This issue affects many-layered networks (feed-forward), as well as recurrent networks.
- Adaptive optimisers can rescale small parameter gradients, but they cannot recover information lost through repeated poorly conditioned Jacobians.

Residual Connections

- One of the most effective ways to make very deep networks trainable is to use residual neural networks (ResNets)[3].
- ResNets are artificial neural networks that use *skip connections* to jump over layers.
- The skip path contributes an identity term to the backward Jacobian, providing a direct route for the gradient.
- This is not the full story: skip connections also change the optimisation geometry and symmetries.

[3] K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, Las Vegas, NV, 2016, pp. 770-778.

Residual Connections

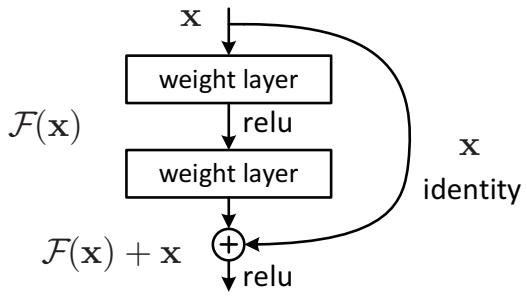


Figure 2. Residual learning: a building block.

K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, Las Vegas, NV, 2016, pp. 770-778.

Residual Gradient Flow

A residual block computes

$$\boldsymbol{y} = \boldsymbol{x} + F(\boldsymbol{x}; \boldsymbol{\theta}) \, .$$

If $J_F(\boldsymbol{x})$ denotes the Jacobian of F evaluated at $\boldsymbol{x}$, then the Jacobian of the block is

$$\frac{\partial \boldsymbol{y}}{\partial \boldsymbol{x}} = I + J_F(\boldsymbol{x}).$$

For a scalar loss L and column-vector gradients, backprop computes $\nabla_{\boldsymbol{\theta}} L$ to update the parameters and $\nabla_{\boldsymbol{x}} L$ to propagate the gradient to earlier layers:

$$\nabla_{\boldsymbol{x}} L = \nabla_{\boldsymbol{y}} L + \underbrace{J_F(\boldsymbol{x})^T \nabla_{\boldsymbol{y}} L}_{\text{VJP through } F} .$$

- The gradient has a direct identity path as well as the path through F.
- This improves gradient flow, although it does not guarantee that gradients remain well scaled.

Residual Connections

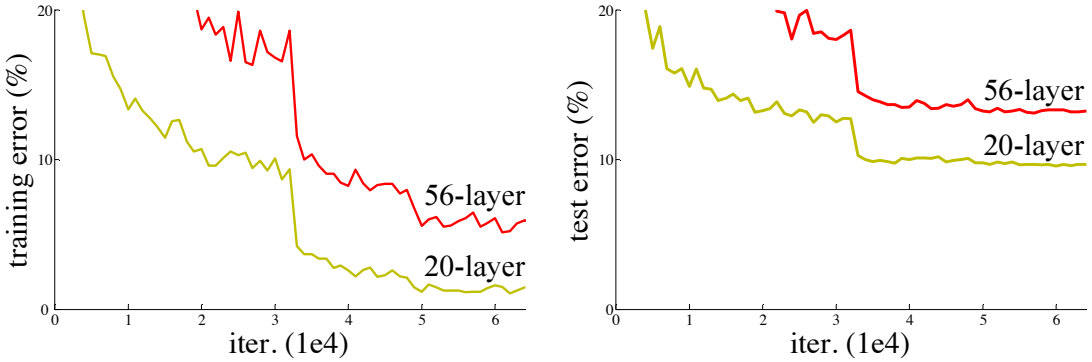


Figure 1. Training error (left) and test error (right) on CIFAR-10 with 20-layer and 56-layer "plain" networks. The deeper network has higher training error, and thus test error. Similar phenomena on ImageNet is presented in Fig. 4.

K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, Las Vegas, NV, 2016, pp. 770-778.

Residual Connections

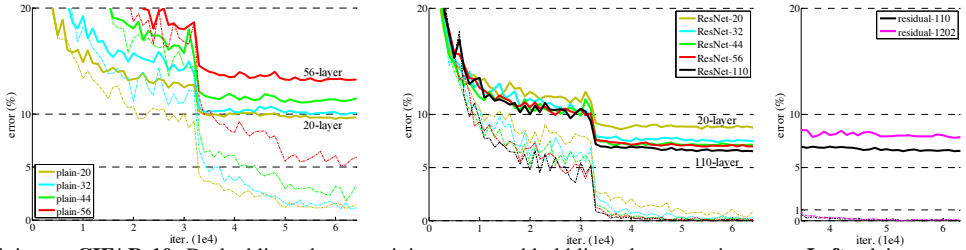


Figure 6. Training on **CIFAR-10**. Dashed lines denote training error, and bold lines denote testing error. **Left**: plain networks. The error of plain-110 is higher than 60% and not displayed. **Middle**: ResNets. **Right**: ResNets with 110 and 1202 layers.

K. He, X. Zhang, S. Ren and J. Sun, "Deep Residual Learning for Image Recognition," CVPR, Las Vegas, NV, 2016, pp. 770-778.

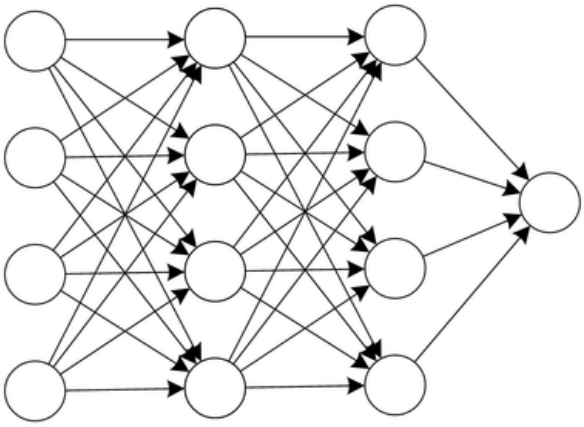
Overfitting

- Neural networks with a large number of parameters (and hidden layers) are powerful, however, overfitting is a serious problem in such systems.
- Just as you've seen in simple machines (e.g. Ridge Regression and LASSO), regularisation can help mitigate overfitting
- In deep networks, we might:
  - Use the architecture to regularise (e.g. ConvNets)
  - Use weight regularisers (L1, L2 [weight decay], etc, ...)
  - Use a stochastic weight regulariser (like dropout)

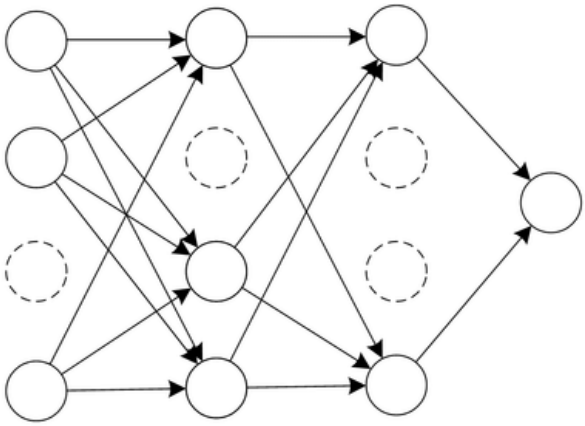
Dropout

- Dropout is a form of regularisation
- The key idea in dropout is to randomly drop neurons, including all of the connections, from the neural network during training.
- Motivation: averaging predictions from several models often improves generalisation.
  - Training many networks independently is expensive.
  - Dropout samples many overlapping subnetworks that share parameters.

Dropout



(a) Standard Neural Network



(b) Network after Dropout

Image from: https://www.researchgate.net/figure/Dropout-neural-network-model-a-is-a-standard-neural-network-b-is-the-same-network_fig3_309206911

How Does Dropout Work?

- In the learning phase, we set a dropout probability for each layer in the network.
- On each training forward pass, we sample a binary mask and apply it to the activations.
- **Inverted Dropout** rescales the retained activations so that their expected magnitude is unchanged when dropout is disabled at evaluation/test time.

How is Inverted Dropout implemented?

- We define a random binary mask $\boldsymbol{m}^{(l)}$ for the activations in layer l. Each element is sampled from a Bernoulli distribution with $P(m_i^{(l)} = 1) = p$, where p is the *keep probability*. A new mask is sampled on each training forward pass.
- In `torch.nn.Dropout`, the argument `p` is instead the *drop probability*, so the keep probability is $1 - p$.
- The forward pass of a Dropout layer (function) during **training** is given by $f(\boldsymbol{x}) = \boldsymbol{x} \odot \boldsymbol{m}/p$.
- The forward pass of a Dropout layer (function) during **inference** is given by $f(\boldsymbol{x}) = \boldsymbol{x}$.
- This can be applied to any layer(s) of the network *except the output layer!*
- It's not common to put it everywhere; just a couple of select places (empirically chosen).
- The gradient (during training) is simply the hadamard product of the incoming gradient with $\boldsymbol{m}/p$.

Why Does Dropout Work?

- A unit cannot rely on a fixed set of other units always being present.
- The noise discourages fragile co-adaptation between features.
- At inference, the full network approximates an average over many overlapping subnetworks.